

Xin YANG

Department of Mechanical Engineering & Division of Mechatronic System Dynamics (LMSD), KU Leuven

CONTACT

- Personal page: <https://xinyang-kul.github.io/>;
- Google Scholar: <https://scholar.google.com/citations?user=Jp41q6AAAAAJ>;
- Work email: xin.yang@kuleuven.be; Personal email: xinyangcqu@gmail.com;
- Building G Room 106, Ghent and Aalst Campuses, Gebroeders De Smetstraat 1, 9000 Gent, Belgium;

EDUCATION

Katholieke Universiteit Leuven

Gent, Belgium

Ph.D. Candidate (22'-), supervised by Prof. Dimitrios Chronopoulos

Jan.2023 - Present

- **Research interests:** Condition monitoring, Multisensory data fusion, Generative AI, Physics-informed NNs
- **PhD thesis:** Multisensory Data Fusion for Aerostructural Damage Assessment: From Configuration Optimization to Hierarchical Fusion;

University of Bristol

Bristol, United Kingdom

Visiting Ph.D. Researcher funded by Research Foundation Flanders (FWO)

Jan.2026 - Feb.2026

- **Affiliation:** Ultrasonics and Non-destructive Testing (UNDT) Group, led by Prof. Bruce Drinkwater;
- **Activities:** Developing landing gear Ultrasonic and X-ray imaging under the supervision of Prof. Anthony John Croxford;

Tongji University

Shanghai, P.R. China

MEng. in Vehicle Engineering

Sep.2019 - Mar.2022

- **Cumulative GPA: 4.67/5.00 (Ranking top 5%);**
- **Scholarships and awards:** Graduate Entry Award, 2019; Outstanding Student Award (2021);
- **Course:** Vehicle dynamics@TJU, Modern Control Theory@TJU, Nonlinear Control Theory@TJU;

Chongqing University

Chongqing, P.R. China

BEng. in Vehicle Engineering

Sep.2015 - Jun.2019

- **Cumulative GPA: 3.68/4.00 (Ranking 5/144);**
- **Scholarships and awards:** National Encouragement Scholarship, 2017; Outstanding Graduate of Chongqing University, 2019 ; Outstanding Student Award (top 3%), 2017 and 2018; Selected to Excellent Student Program (top 5%, on basis of outstanding academic performance);

PROJECTS

- [1] **FWO Visiting Grant** (PI, 1650 €, No.K200826N): FWO-funded visiting grant to conduct a research visit at the University of Bristol, [Ultrasonics and Non-destructive Testing \(UNDT\)](#) group.
- [2] **FWO-Poland Weave Project** (No.G039226N): involved in the FWO-Poland Weave project, collaborated with Prof. [Anna Boczkowska](#) from Warsaw University of Technology.

JOURNAL PAPERS

- [1] **Yang, X.***, Fang, C., Liao, Y. L., Yang, J., Gryllias, K., Chronopoulos, D. (2026). Deep Generative Models in Condition and Structural Health Monitoring: Opportunities, Limitations and Future Outlook. *Mechanical Systems and Signal Processing*, 253, 114277. doi:10.1016/j.ymssp.2026.114277. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [2] **Yang, X.***, Cantero-Chinchilla, S., Moradi, M., Komninos, P., Fang, C., Liao, Y.L., Kundu, P., Zarouchas, D., Chronopoulos, D. (2025). Damage imaging in structural health monitoring with fine-tuned conditional diffusion model. *Mechanical Systems and Signal Processing*, 236, 112996. doi:10.1016/j.ymssp.2025.112996. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [3] **Yang, X.***, Fang, C., Kundu, P., Yang, J., Chronopoulos, D. (2024). A decision-level sensor fusion scheme integrating ultrasonic guided wave and vibration measurements for damage identification. *Mechanical Systems and Signal Processing*, 219, 111597. doi:10.1016/j.ymssp.2024.111597. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [4] **Yang, X.***, Farrokhhabadi, A., Rauf, A., Liu, Y. C., Talemi, R., Kundu, P., Chronopoulos, D. (2024). Transfer learning-based Gaussian process classification for lattice structure damage detection. *Measurement*, 238, 115387. doi:10.1016/j.measurement.2024.115387. (JCRQ1, IF: 5.6) [\[paper\]](#)
- [5] Vu, N.V.H., Yang, J., Kiran, R., **Yang, X.**, Thanh, P.T. (2026). Effect of discontinuous elastic foundations on geometrically nonlinear bending and stress response of thick composite plates via energy-based physics-informed neural networks. *Thin-Walled Structures*, 226, 114876. doi:10.1016/j.tws.2026.114873. (JCRQ1, IF: 6.6) [\[paper\]](#)
- [6] Yan, T.T, Xing, X.Q., Mo, Z.L., **Yang, X.**, Wang, D., Tsui, K.L. (2026). Integration of Dynamic Time-Varying Weighting and Interpretable Optimization Modeling for Practical Degradation Stage Assessment. *Reliability Engineering & System Safety*, 272, 112515. doi:10.1016/j.res.2026.112515. (JCRQ1, IF: 11) [\[paper\]](#)
- [7] Du, J. B., Zhao, H. H., **Yang, X.**, Cui, J. L. (2026). Impact behavior of laminates: Role of energy gradients and layup. *International Journal of Mechanical Sciences*, 312, 111274. doi:10.1016/j.ijmecsci.2026.111274. (JCRQ1, IF: 9.4) [\[paper\]](#)
- [8] Fang, C., **Yang, X.**, Shi, S.M., Gryllias, K., Vandepitte, D., Liu, X., Zhang, L., Chronopoulos, D. (2026). Bayesian Waveform Inversion for Structural Parameter Estimation Using Narrow Spectral Characteristics of Guided Waves. *Mechanical Systems and Signal Processing*, 244, 113827. doi:10.1016/j.ymssp.2025.113827. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [9] Liao, Y. L., Wang, Y. H., Fang, C., **Yang, X.**, Zheng, X. P., Chronopoulos, D., Qing, X. L. (2025). Baseline-free damage detection and localization on composite structures with unsupervised Kolmogorov-Arnold autoencoder and guided waves. *Mechanical Systems and Signal Processing*, 239, 113356. doi:10.1016/j.ymssp.2025.113356. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [10] Fang, C., **Yang, X.**, Gryllias, K., Vandepitte, D., Liu, X., Zhang, L., Chronopoulos, D. (2025). Using removable sensors in structural health monitoring: A Bayesian methodology for attachment-to-attachment uncertainty quantification. *Mechanical Systems and Signal Processing*, 224, 111973. doi:10.1016/j.ymssp.2024.111973. (JCRQ1, IF: 8.9) [\[paper\]](#)
- [11] Mardanshahi, A.* , Sreekumar, A., **Yang, X.***, Barman, S.K., Chronopoulos, D. (2025). Sensing Techniques for Structural Health Monitoring: A State-of-the-Art Review on Performance Criteria and New-Generation Technologies. *Sensors*, 25(5), 1424. doi:10.3390/s25051424. (JCRQ1, IF: 3.5) [\[paper\]](#)

- [12] Lu, H., Chinchilla, S., **Yang, X.***, Gryllias, K., Chronopoulos, D. (2024). Deep learning uncertainty quantification for ultrasonic damage identification in composite structures. *Composite Structures*, 338, 118087. doi:10.1016/j.compstruct.2024.118087. (JCRQ1, IF: 7.1)[[paper](#)]
- [13] Farrokhhabadi, A., Lu, H., **Yang, X.**, Rauf, A., Talemi, R., Chronopoulos, D. (2024). Energy Absorption Assessment of Recovered Shapes in 3D-Printed Star Hourglass Honeycombs: Experimental and Numerical Approaches. *Composite Structures*, 347, 118444. doi:10.1016/j.compstruct.2024.118444. (JCRQ1, IF: 7.1)[[paper](#)]
- [14] Hu, X., **Yang, X.**, Feng, F., Liu, K., Lin, X. (2021). A particle filter and long short-term memory fusion technique for lithium-ion battery remaining useful life prediction. *Journal of Dynamic Systems, Measurement, and Control*, 143 (6), 061001. doi:10.1115/1.4049234. (JCRQ2, IF: 1.3)[[paper](#)]

CONFERENCE PAPERS

- [1] **Yang, X.**, Kundu, P., Chronopoulos, D. (2025). Fine-tuned Diffusion Model for High-resolution Sparse-Array Ultrasonic Guided Wave Imaging. *The 15th Prognostics and System Health Management Conference (PHM 2025)*. [[paper](#)]
- [2] **Yang, X.**, Chen, F. (2022). Deep Uncertainty Quantification of Prognostic Techniques for Proton Exchange Membrane Fuel Cell. *Vehicle Electrification and Powertrain Diversification Technology Forum Part II*. [[paper](#)]
- [3] **Yang, X.**, Chen, F., Jiao, J., Liu, S. (2021). Machine learning-based voltage degradation prediction with uncertainty quantification for PEMFC. *2021 5th CAA International Conference on Vehicular Control and Intelligence (CVCI)*. [[paper](#)]

ACADEMIC ACTIVITIES

- Journal Reviewer:
 - Mechanical Systems and Signal Processing
 - IEEE Trans on Industrial Informatics
 - IEEE Trans on Industrial Electronics
 - IEEE Trans on Instrumentation and Measurement
 - Expert Systems with Applications
 - Engineering Applications of Artificial Intelligence
 - Advanced Engineering Informatics
 - Reliability Engineering & System Safety

WORKING EXPERIENCE

- | | |
|---|---|
| <p>NIO Inc.</p> <p><i>Intern @ Vehicle Chassis Control Group</i></p> <ul style="list-style-type: none"> • Lane centering control, Hardware in-the-loop testing • Ethernet/CAN communication protocol | <p>Shanghai, China</p> <p><i>Feb. 2022 - Jun.2022</i></p> |
| <p>Holomatic Inc.</p> <p><i>Intern @ Perception Group for Autonomous Driving</i></p> <ul style="list-style-type: none"> • 3D object detection, Driving lane detection/segmentation • Remote server model training, git | <p>Shanghai, China</p> <p><i>Sep. 2021 - Nov.2021</i></p> |

OTHER SKILLS



Coding: Python (Pytorch), MATLAB (Simulink), C++/C (LeetCode 300+), Linux(Ubuntu), and etc.



Languages: Mandarin, English (IELTS 7.0), Dutch (A2)



Expertise: Generative AI, Deep Learning, Object detection, Defect segmentation, Data Analysis & Learning, MATLAB Simulink System & Control, and etc.